Retrieval-Augmented Question Answering with Knowledge Graphs[1]

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Abstract

Retrieval-Augmented Generation (RAG) has seen rapid advancements in recent years, leveraging external sources of information to enhance language model outputs. Knowledge Graphs (KGs), which store structured, interpretable information, offer a promising avenue for improving retrieval in RAG systems. In this work, we explore the use of KGs to enhance question answering on the PubMedQA dataset, a benchmark consisting of yes/no/-maybe questions about biomedical research articles from the PubMed database. While PubMedQA is typically used to evaluate LLM reasoning on isolated questions, we propose a novel approach that integrates KG-assisted retrieval into the RAG pipeline. By designing dataset-specific prompting strategies, we extracted key scientific entities and relations from biomedical abstracts to construct a tailored KG that links relevant contextual information to each question. Although our knowledge graph–based system (57% accuracy) did not outperform a vector database (77% accuracy), we found that the structure of the KG and our prompting techniques had a measurable impact on model accuracy. These findings highlight the potential of combining structured knowledge representations with carefully crafted prompts to improve domain-specific question answering. Additional experimentation demonstrated that as the domain of the knowledge bases become more narrow, knowledge graphs perform better when compared to vector databases.

1 Introduction & Relevant Literature

1.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) architecture is an AI framework that uses information retrieval from external documents to assist in response generation. It allows a model to generate responses with relevant information without retraining the model (Lewis et al., 2020).

To enhance retrieval efficiency, vector-based models have become essential components of modern RAG implementations. A RAG vector database retrieves relevant information for a large language model (LLM) by converting documents into vector embeddings and matching a user query with the most similar vectors (Berry et al., 1999). This approach is ideal for unstructured data, but it may not be the most effective method for capturing the schema and relationships inherent in structured information.

1.2 Knowledge Graph Retrieval-Augmented Generation

Researchers have started integrating knowledge graphs into RAG systems to further improve knowledge retrieval, reasoning, disambiguation, and factual consistency. A knowledge graph (KG) is a structured representation of named entities (e.g. people, places, organizations, concepts) and their relationships, forming a graph-based knowledge structure for organizing and retrieving data. While traditional RAG models retrieve unstructured data, KG-based RAG instead retrieves structured knowledge from graph databases. The integration of KGs into the RAG framework improves the relevant information retrieval performance, especially in domain-specific use cases (Zhong et al., 2023; Pan et al., 2024).

1.3 Automatic Knowledge Graph Construction (KGC)

Constructing a knowledge graph from a corpus is a complex process that requires structuring information into meaningful relationships. Knowledge graphs are often built from scratch by domain experts, derived from unstructured or semi-structured data, or integrated from existing knowledge graphs. This process typically involves entity extraction,

[1]Code available at https://github.com/jaulie/knowledge_graph_creation

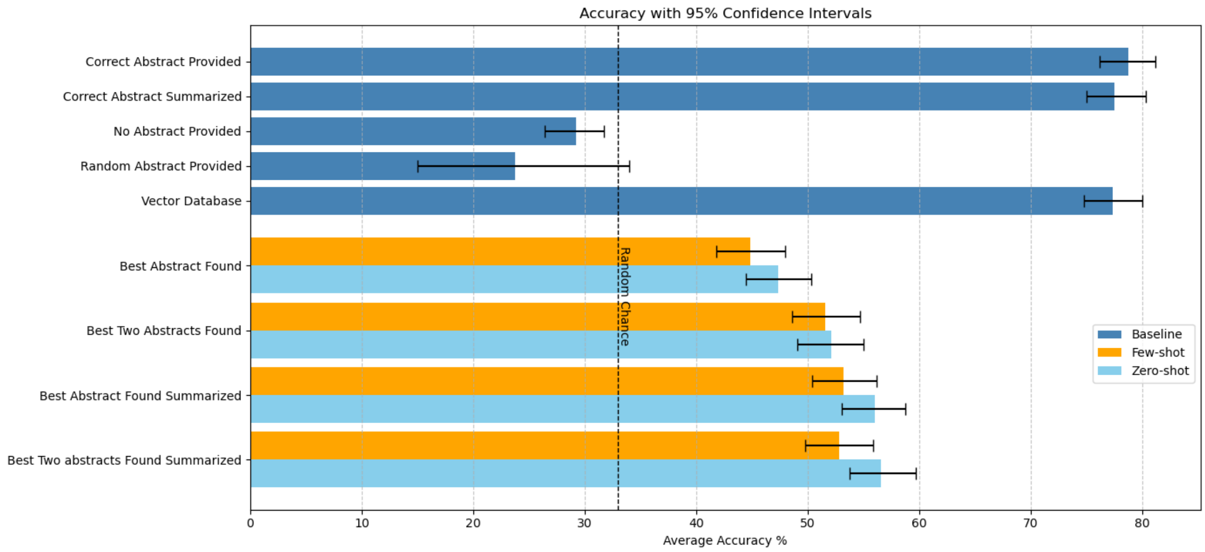

Figure 1: We evaluated answer accuracy across four baselines—perfect, no, random, and question-only retrieval—and compared these to a vector database system. We also report results from eight ablation variants, isolating the effects of key system components. Together, these experiments offer a comprehensive view of how retrieval quality and system design influence performance.

relationship identification, and ontology development, which can require extensive human annotation and specialized expertise (Zhong et al., 2023).

Recent advancements in LLMs have made it possible not only to automate the process of KGC, but also to build sequence-to-sequence systems that perform all subtasks necessary to construct KGs. Universal Information Extraction System is an early example that demonstrated the feasibility of this approach (Lu et al., 2022). Early KGC systems fine-tuned pre-trained language models on information extraction subtasks, breaking the overall challenge into parts such as entity recognition, coreference resolution, and relation extraction, then combining those subtasks into an overall system (Wang et al., 2020; Melnyk et al., 2022; Dognin et al., 2021).

1.4 KGC with Prompt Engineering

As pre-trained language models have increased in size, so has their potential for KGC through zero- and few-shot prompting. Compared to smaller pre-trained models that required task-specific fine-tuning, current state-of-the-art LLMs demonstrate remarkable latent capabilities for performing novel tasks. Prompting is a reliable alternative to fine-tuning for generation (Qian et al., 2022), but LLMs are sensitive to prompt engineering (Wei et al., 2023). Prompt-tuning has been used for upstream tasks, such as relation extraction (Chen et al., 2021) and event detection (Wang et al., 2022).

Prompting-based methods have been used for sequence-to-sequence KGC (Zhang and Soh, 2024; Zhu et al., 2024) and for the similar task of relational triple extraction (Ding et al., 2024). For instance, (Carta et al., 2023) introduced a pipeline leveraging GPT-3.5 to construct knowledge graphs through iterative zero-shot prompting. Each prompt operates without examples, allowing the LLM to autonomously extract relevant entities and relationships from unstructured text with no manual intervention. The pipeline systematically identifies entities, assigns types and descriptions, extracts relationships, resolves duplicates, and infers a schema, ensuring a structured and coherent knowledge graph.

1.5 PubMedQA Dataset

PubMedQA is a dataset containing medical questions and corresponding yes/no/maybe answers, along with relevant context (Jin et al., 2019). This dataset builds on top of the PubMed dataset (Public Biomedical Literature library), and each question in this dataset specifically corresponds to one paper in the PubMed dataset. This dataset is traditionally used to evaluate a model's reasoning capabilities on a question-by-question basis.

1.6 Novel Component of Our Experiment

There have been uses of iteratively prompting an LLM to build knowledge graphs. These have had a variety of approaches to model evaluation, such as comparing steps of the pipeline to baseline models for specific tasks (Zhang and Soh, 2024), employing human assessors to evaluate the quality of retrieved documents (Carta et al., 2023), or evaluating on reasoning benchmark datasets as opposed to retrieval benchmark datasets (Wei et al., 2023). To date, there has not been research specifically evaluating the document retrieval capabilities of a RAG model using a KG constructed through LLM prompting, assessed against a related ground-truth QA dataset.

1.7 Study Rationale

In the medical domain, the use of incorrect or outdated information can have catastrophic consequences. Therefore, any model that provides answers to medical questions must use relevant and up-to-date information in response generation, with explainable reasoning steps. This is where a RAG model with a KG database can be useful.

1.8 Hypothesis

We propose that a variety of prompting techniques can enable LLMs to generate richer knowledge graphs that enhance information retrieval in RAG systems more effectively than traditional vector database approaches.

1.9 Summary of Results

While our KG-RAG model underperforms relative to the vector database baseline on the full PubMedQA dataset, further experimentation indicates that knowledge graph-based retrieval can be reasonably effective in settings where documents are topically cohesive and share overlapping named entities.

2 Methodology

2.1 RAG Pipeline

Our experimental design for knowledge graph creation involved testing systems that automatically generate knowledge graphs using various prompting methods. Our base model for KGC was LlaMA 3.2.

The resulting KGs were incorporated into a RAG pipeline using LangChain, a framework for developing applications powered by language models, and Neo4j, a graph database management system. Once in Neo4j, an LLM assisted in converting a given question to a Cypher query, which was then used to retrieve relevant information from the KG to answer the question. Cypher is a graph query language that allows for effective querying of property graph databases.

Performance was measured by question-answering accuracy on the set of PubMed abstracts associated with the labeled questions in the PubMedQA dataset. This dataset is traditionally used for evaluating reasoning capabilities given an individual abstract. However, we use it to evaluate retrieval across many abstracts. Domain-specific contexts are a key use case for KG-RAG, as state-of-the-art LLMs often struggle with insufficient domain knowledge (Ling et al., 2024). Domain-specific fine-tuning and pre-training are cost prohibitive, which make knowledge graphs a practical alternative in these scenarios.

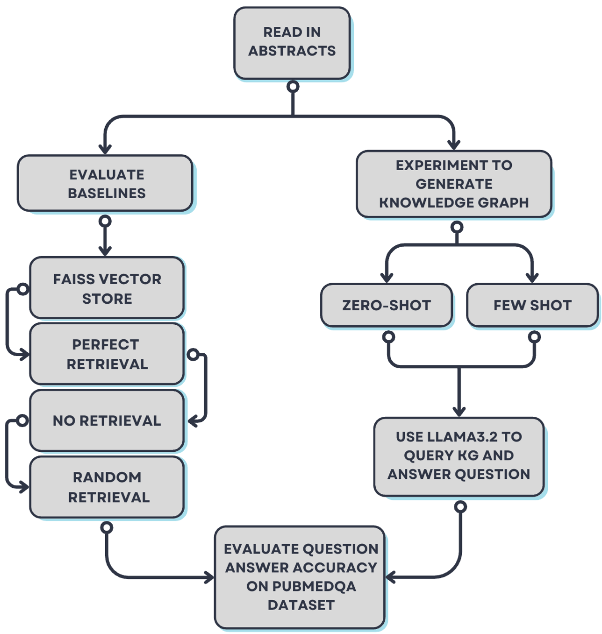

Figure 2: Experiment Overview

2.2 Prompt Engineering

For our experiments, prompt design serves as the primary hyperparameter guiding model performance. We experimented with iterative refinement of LLM prompts, testing different strategies for extracting more granular relationships and improving the quality of the knowledge graph. This method of determining the optimal prompt involved multi-step iterative prompting, when an initial extraction pass is refined through subsequent interactions. We experimented with zero-shot prompting, with no examples to aid the model, as well as few-shot prompting, where relevant examples guide the model's relationship extraction process. In the context of medical question answering, the examples used in few-shot prompting were tailored to align with how the dataset expects the model to select information.

2.3 Dataset-Specific Decisions

In the PubMedQA dataset, abstracts are divided into distinct sections: the *Title* which is converted to a question to support question-answering tasks; the *Context*, which includes the abstract content excluding the conclusion; the *Long Answer* which corresponds to the conclusion; and *Ground Truth*, which indicates the correct answer as "Yes," "No," or "Maybe" (Jin et al., 2019).

In our pipeline, we extracted entities from the body of each abstract, excluding the title, which serves as the question for evaluation. The extracted scientific terms are treated as entities in our knowledge graph, each of which is linked back to its corresponding source paper. Within the RAG framework, we extracted entities from each question, used the knowledge graph to retrieve relevant abstracts, and provided these as context for the LLM to generate an answer.

2.4 Evaluation

As noted earlier, we evaluated our model using the PubMedQA dataset, where each question must be answered with one of three options: yes, no, or maybe. Model performance was measured by accuracy, calculated as the proportion of correct responses compared to the labeled ground-truth answers. The PubMedQA dataset contains 1,000 QA instances labeled by experts, as well as additional instances that are either unlabeled or generated by AI. For our evaluation, we use only the expert-labeled subset, as accurate performance assessment requires questions with known ground-truth answers. Consequently, the unlabeled and AI-generated instances are excluded from this study.

RAG pipelines often employ vector databases as document stores, using cosine similarity to retrieve relevant documents. Since this is the current state-of-the-art in RAG systems, we test our system against a baseline where a vector database is used instead of a knowledge graph to store documents.

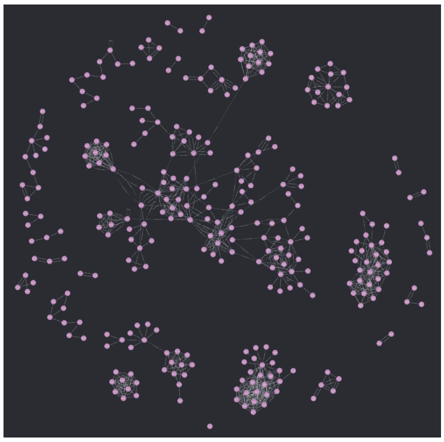

Figure 3: Example of structured knowledge graph generated by extracting simple relations from 50 labeled examples in the PubMedQA dataset. Resulting graph is sparse, with many isolated nodes and small clusters, reflecting the limited entity overlap across individual abstracts.

3 Experiments

3.1 Initial Experiments

In our initial experiments, we used LLaMA 3.2 to extract sentence-level relations like:

(Gastroenteropancreatic Neuroendocrine Tumors) —[:IMPACT]→ (Sst2A Immunohistochemistry)

This produced highly varied relationships that were difficult to use when answering the questions. We then restricted the relationships to indicate only positive or negative associations between entities, but this linking was too limited to capture the nuances of the questions. Next, we restricted the output to a set of 30 predefined relationships commonly used to answer the questions, but this approach still lacked sufficient flexibility.

We realized that it would be more effective to store the key elements of each abstract in the knowledge graph so that we could identify which abstract was relevant based on the ideas presented in the question. In our later approach, we found that allowing the RAG model to use the full associated abstract as context was much more effective than limiting the answer to the relationships found in the knowledge graph.

3.2 Main Experiment

Our new approach involved storing the abstract-specific Article ID alongside each key entity extracted from an abstract within the KG. Each full abstract was also linked to its corresponding Article ID in the KG. To populate the graph, we used both zero- and few-shot prompting strategies to extract key entities from the abstracts. During inference, we extracted entities from the question and applied fuzzy matching to compare them with entities in the KG. The abstract associated with the highest number of matched entities was then selected as context for the LLM.

Using the top-matching abstract(s), we prompted LlaMA 3.2 to answer the question. In one variation, we first asked the model to summarize the conclusion of the retrieved abstract and then provided both the summary and the question as input to the LLM. In another variation, the two top-matching abstracts were combined with the question to generate an answer, rather than relying on a single abstract.

Our ablation study tested three main variables:

- **Prompt type:** zero-shot vs. few-shot
- **Content type:** full abstract vs. summary
- **Number of abstracts retrieved:** top 1 vs. top 2

These factors resulted in eight different experimental configurations.

4 Results

4.1 Baseline Results

Vector Database Retrieval (FAISS Baseline). To compare the effectiveness of different retrieval strategies, we implemented the LangChain FAISS vector store as the retrieval mechanism for a RAG system, using a vector database in place of the knowledge graph. The vector index was initialized with embeddings of the context and conclusions from labeled PubMedQA questions. At inference time, the input question was embedded, and the most semantically similar abstracts were retrieved based on vector similarity. This context was then passed to the LLM for answer generation.

Controlled Retrieval Conditions. To establish meaningful performance bounds, we evaluated LLaMA 3.2 under three controlled retrieval conditions:

- **Perfect Retrieval**: The model was given the ground-truth abstract corresponding to each question, representing the ideal upper-bound for performance.
- **No Retrieval**: The model was prompted with only the question, without any supporting abstract, to measure its baseline reasoning ability in isolation.
- **Random Retrieval**: The model received a randomly selected abstract to simulate poor retrieval and assess sensitivity to irrelevant or misleading context.

These settings, along with our RAG baselines using vector and knowledge graph retrieval, provide a comprehensive framework for evaluating how retrieval quality impacts downstream performance.

Baseline Comparisons and Observations. We evaluated several baselines to contextualize our system's performance. With three possible answer choices, random guessing would yield an expected accuracy of approximately 33.3%. Surprisingly, in the no retrieval condition, the model performed below chance, underscoring its limited ability to answer questions without relevant context. Performance further deteriorated in the random retrieval setting, where unrelated abstracts introduced noise that not only misled the model but also produced a wider confidence interval, reflecting high variability across trials.

Upper Bound and Vector Baseline Results. The perfect retrieval condition achieved the highest accuracy at 78.7%, representing the upper limit of model performance under ideal context. Interestingly, summarizing the ground-truth abstract before providing it to the model slightly reduced accuracy, likely due to loss of fine-grained information.

The FAISS vector database baseline performed just below this upper bound, achieving 77% accuracy. This result highlights the effectiveness of vector-based semantic retrieval, which, despite lacking perfect precision, offers robust and efficient context retrieval for question-answering.

4.2 Novel KG-RAG Pipeline Results

The zero-shot approach unexpectedly outperformed the few-shot method across all configurations. While counterintuitive, we believe improved few-shot examples could eventually reverse this.

Using zero-shot prompting, our system correctly retrieved the relevant abstract 40.9% of the time. When the second-best match was also considered, this increased to 46.6%. Based on the retrieval frequency of the correct abstract and the model's known performance with and without it, we estimated a chance-level accuracy of 46.4% for the knowledge graph setup using a single retrieved abstract. Our system achieved an actual accuracy of 47.3%, slightly exceeding this baseline. This suggests that even when the top retrieval was not the correct abstract, the retrieved content was still rele-

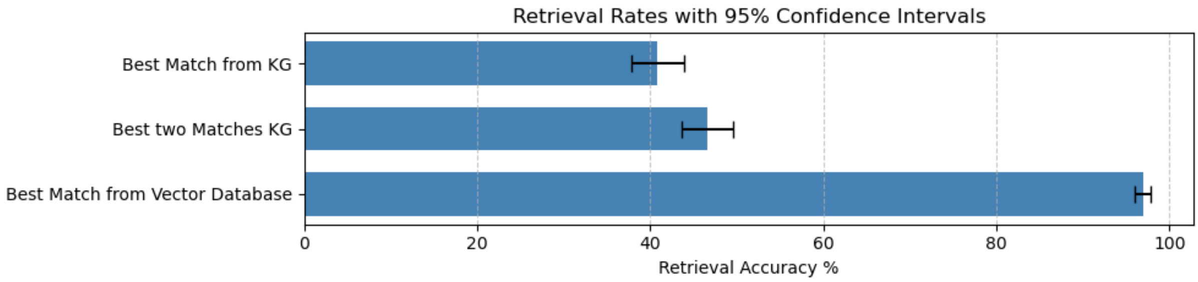

Figure 4: Retrieval accuracy rates with 95% confidence intervals for knowledge graph (KG) and vector database approaches. The vector database consistently retrieves the correct abstract with higher accuracy than both the best match and top-two matches from the knowledge graph. KG-based retrieval shows moderate performance with substantial variability across examples, reflecting challenges in entity alignment and sparse graph connectivity.

vant enough to aid the model's response. However, the improvement is modest, and further testing is needed to determine whether it reflects a true advantage or falls within the margin of error.

Although summarizing the correct abstract reduced accuracy slightly, it seemed to help when the retrieved abstract was only partially relevant. This improvement was large enough; it appears very likely that summarizing the abstracts helped arrive at the correct answer more often than would be expected with a randomly selected abstract. We hypothesize that summarization filtered out extraneous details, making it easier for the model to focus on relevant content.

When both the top and second-best abstracts were used, combined with summarization, the marginal gains shrank and fell within the 95% confidence interval, suggesting the improvement may be due to chance.

The experimental methods described in this paper had a quantifiable positive impact on the performance of the RAG model, with zero-shot with a summary of the best abstract performing the best. However, despite various efforts to improve accuracy through careful entity extraction and prompt engineering, the vector database ultimately outperformed the knowledge graph in our experiments.

5 Discussion

The biggest area for improvement is in entity extraction. While most extracted relationships are meaningful, there are cases where the LLM fails to produce them in the correct format. There are also meta-relationships or relationships requiring world knowledge that are not extracted by the LLM. This limits the amount of context retrieved and ultimately reduces accuracy. Refining this step could improve both the zero-shot and few-shot approaches, with a potentially greater effect on the few-shot configuration. A closer review of the relationships generated across all abstracts could help identify common failure patterns and guide refinements. Better entity extraction would directly improve retrieval quality and lead to more accurate model responses.

The creation and use of the knowledge graph were substantially more compute-intensive than working with a standard vector database. Processing all 1,000 abstracts, including summary generation and entity extraction, took approximately 4.5 hours, whereas the vector model completed its processing in under 30 minutes.

Given that KG-based approaches also produced lower accuracy, they do not currently offer a strong alternative to vector-based retrieval. Searching across more abstracts increases the likelihood of retrieving documents that are similar but not correct. More testing is needed to understand how this affects retrieval performance.

The success of the vector database suggests that, in the current setup, semantic similarity-based retrieval offered more reliable context for QA than the structured but potentially brittle reasoning supported by the KG. These results point to the need for further refinement in knowledge graph construction and utilization, particularly in how entities and relations are selected and represented.

6 Further Experiments

6.1 Motivation

To better understand the performance gap between knowledge graphs and vector databases, we conducted a focused experiment on a smaller, domain-specific subset of the PubMedQA dataset. This

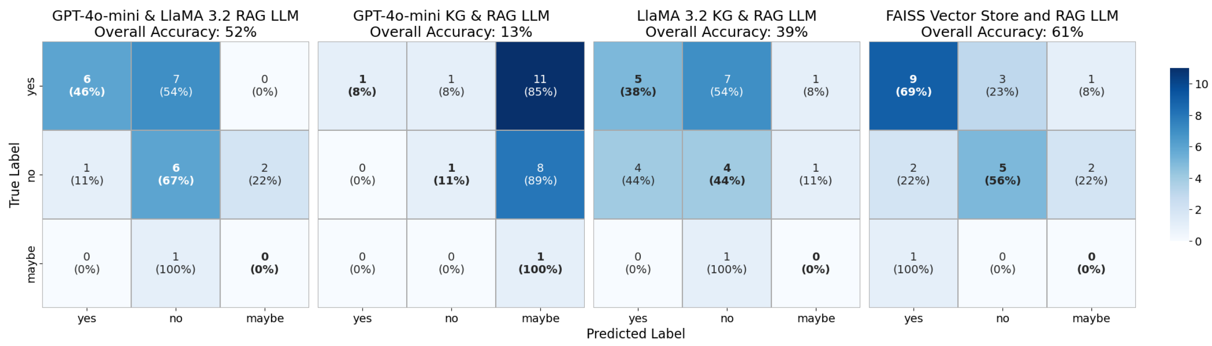

Figure 5: Confusion matrices comparing the performance of different RAG-based LLM systems on a domain-specific subset of 23 PubMedQA questions related to diabetes. Each matrix shows raw counts and percentage distributions across answer classes ("yes", "no", "maybe"), with bolded diagonals indicating correct predictions. Overall accuracy is reported above each matrix. The FAISS vector database combined with an LLM achieves the highest accuracy (61%), outperforming knowledge graph–augmented variants using GPT-4o-mini and LLaMA 3.2.

subset consisted of 23 questions and abstracts, all related to diabetes. Our hypothesis was that a more topically cohesive dataset might benefit more from a knowledge graph-based approach, especially in cases where semantic search could struggle to differentiate between closely related documents.

6.2 Methods

We built two versions of the knowledge graph using different large language models. The first was created using LLaMA 3.2, which extracted entities and relations and then refined the relation set, reducing the number of unique relations from 73 to 64. This graph contained 182 nodes, 174 relations, and 26 connected components, with a density of 0.0106. The second graph was built using GPT-4o-mini in an attempt to extract more fine-grained or denser relationships. The resulting graph had 331 nodes, 307 relations, and 43 connected components, with 145 unique relation types and a density of 0.0056.

We tested three configurations:

- **1**: GPT-4o-mini for both graph creation and querying
- **2**: LLaMA 3.2 for both graph creation and querying
- **3**: GPT-4o-mini for graph creation and LLaMA 3.2 for querying

In each case, we extracted the main entities from the question and used the set of outgoing relations corresponding to those entities as input to the language model.

As a baseline, we applied the same vector database retrieval approach used previously, leveraging the FAISS vector store. Abstracts in the diabetes subset were embedded using the same sentence-transformer model, and queries were matched against this smaller database to retrieve top-ranked documents for question-answering.

6.3 Results

GPT-4o-mini, when used as the answering model, performed poorly and frequently defaulted to the answer "maybe." The best performance came from the configuration where GPT-4o-mini was used to build the knowledge graph and LLaMA 3.2 was used to query it. This system achieved an accuracy of 52.1%.

For comparison, the vector database approach achieved 61% accuracy on the diabetes subset, though its performance was higher on the full dataset, where accuracy reached 77%. The drop in vector performance on the smaller subset supports our hypothesis that semantic search may struggle when many documents contain similar content. However, even with a denser and more focused knowledge graph, the KG-based approach did not outperform the vector database.

6.4 Discussion

One possible reason the vector database continues to outperform is that it benefits from flexible, surface-level semantic matching. This allows it to retrieve relevant information even when the phrasing or terminology varies. In contrast, the knowledge graph may suffer from sparsity, incomplete entity linking, or brittle graph traversal logic, all of which limit its ability to surface the most useful context.

Beyond the limitations of the model and the graph, the characteristics of the PubMedQA dataset may play a role. Each question is associated with a single abstract, and there is minimal overlap between entities across documents. As a result, semantic search is often sufficient, since retrieving the correct document usually guarantees access to the relevant context. This favors vector-based approaches, which can return the correct document with high precision and speed.

In contrast, knowledge graphs are more useful in domains where documents are richly interconnected and multi-hop reasoning is required. While our high-level schema improved performance, it still fell short of matching the efficiency and accuracy of the vector database.

7 Conclusion

We implemented a KG-RAG model to improve information retrieval for medical question-answering, and evaluated it using the PubMedQA dataset. We used a variety of prompting techniques for entity extraction, mapping scientific terms to related information. In the broad domain of biomedical data, this model underperforms when compared to a vector database implemented in a similar RAG model. Further experimentation indicated that the KG-RAG model performs better in domains that are more specific, with scientific terms that overlap with more frequency.

7.1 Ethical Concerns

The use of LLMs for advice raises ethical concerns about accountability, transparency, and bias. These concerns are amplified in the field of medicine, where incorrect advice can lead to catastrophic outcomes. Thus, the scope of the use of a RAG model for medical questions is important to identify. Decisions regarding its use depend on the requirements and benchmarks set by professionals in the medical field.

7.2 Future Work

This experiment highlights several areas for improvement. Larger and more topically linked graphs performed better and were closer to baseline accuracy, suggesting that improving graph connectivity is crucial. A more advanced querying strategy that supports multi-hop reasoning and filters out extraneous context may also improve results, as irrelevant information often misguides the model.

There are several promising directions for refining knowledge graph construction. These include pre-chunking documents before entity extraction, using iterative prompting to improve triple quality, and applying summarization to increase node connectivity. Information loss remains a major challenge in knowledge graph creation, but in some cases, it may be offset by gains in structure and reasoning capability. Further work is needed to determine whether this tradeoff is worthwhile, especially in datasets better suited to graph-based reasoning.

7.3 Division of Labor

Throughout the course of this project, each team member contributed in distinct and meaningful ways. Jaulie developed the initial RAG pipeline using Neo4j and LLaMA 3.2 on a dummy movie dataset, which served as a proof of concept for later experiments on the PubMedQA dataset. She later worked with a focused subset of diabetes-related abstracts and experimented with the default LangChain knowledge graph implementation to establish a lower-bound baseline for KG-based retrieval. Ryan explored various techniques for knowledge graph generation, focusing on designing a more streamlined and interpretable relation schema that could generalize across abstracts. Brian worked on building a more robust and fine-grained knowledge graph representation, leading to a team-wide pivot toward a new structure for the RAG pipeline, which he then implemented. Brian also conducted a series of experiments testing different prompting techniques to further optimize the system. Rebecca developed the vector database baseline using FAISS and curated the smaller subset of diabetes-related abstracts used in later experiments. She also discovered that the existing file structure in the PubMedQA dataset could be repurposed for entity and relation extraction, eliminating the need to manually download abstracts from PubMed. All team members contributed to writing and revising the final report.

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